

Machine learning fascinates me because it translates data into decisions that matter. What began as curiosity about how data can drive decisions has grown into a passion for building intelligent systems. Through projects, competitive programming, and software development, I have explored different aspects of machine learning and AI, and I am applying to Amazon ML Summer School 2026 to strengthen my foundations under expert mentorship.

I am pursuing a B.E. in Computer Science Engineering at St. Joseph's College of Engineering, Chennai, with a **CGPA** of **9.14**. Alongside academics, I have focused on developing strong problem-solving skills through data structures, algorithms, and software development, while exploring machine learning through independent learning and hands-on implementation.

In my **Water Depletion Prediction System**, I worked on synthetic data generation, exploratory data analysis, feature engineering, and developed a Random Forest Regressor achieving an R^2 score of 0.967 with strong cross-validation performance. Building and publishing the dataset and model on Kaggle taught me the importance of reproducibility and careful experimentation in ML workflows.

I further explored applied machine learning through **HireReady**, an end-to-end application that predicts student job readiness using academic and skill-based features. Developing this project involved data preparation, feature engineering, model training, and deployment through an interactive interface. More importantly, it taught me that machine learning is valuable only when its predictions can be translated into actionable insights for users.

Beyond traditional ML, I explored large language models through a **Retrieval-Augmented Generation (RAG)** prototype. I built a system that processes PDF documents, generates semantic embeddings using Hugging Face models, stores them in a Chroma vector database, and retrieves relevant context for Gemini-based responses. Working on this project introduced me to information retrieval, vector search, and the importance of grounding LLM outputs with reliable external knowledge.

My experiences outside machine learning have also shaped my approach to problem solving. As a Tech Intern at **Xebia**, I gained exposure to enterprise application development, professional SDLC practices, and teamwork. Leading integration efforts for **RoadSOS** during the IIT Road Safety Hackathon further developed my ability to coordinate technical work, manage GitHub workflows, and deliver a functional MVP under time constraints.

Beyond projects, **competitive programming** has shaped the way I think. Regular participation across platforms such as LeetCode, CodeChef, Codeforces, and AtCoder has strengthened my algorithmic reasoning, persistence, and ability to break down complex problems into structured solutions.

What draws me to **Amazon ML Summer School** is the opportunity to learn from experts working on machine learning at scale while interacting with peers who share the similar enthusiasm for technology. I want to deepen my understanding of machine learning fundamentals and gain insights into designing robust, real-world AI systems.

I believe the best way to learn is by building, experimenting, and continuously improving. Through my academic journey and experiences, I have developed the curiosity, discipline, and problem-solving mindset needed to make the most of this opportunity. I am eager to learn, collaborate, and strengthen my foundations so that I can build machine learning solutions with real-world impact.